PROJEKT MANAGEMENT AUSTRIA

member of IPMA

Project handbook

Smart-Wand

1

Version 7

Project manager: Thomas Toferer

Date: 16/12/25

Content

Document versions

Versionno.	Date	Change	Author
1	17.11.2025	Initial Entries	Toferer
2	1.12.2025	Finalising initial Entries	Durkowitsch
2.5	1.12.2025	Creating Project Environment Analysis	Codoban
3	2.12.2025	Finishing the Parts for Project Handbook Part 1 up to 1.10	Toferer, Durkowitsch
4	13.12.2025	Updating Distribution list, Envirement Graphic, WBS,	Toferer, Durkowitsch, Codoban
5	2.2.2026	Updating PSP, Filling out the rest, Adding progress report	Toferer
6	8.3.2026	Updating PSP, Project Progress Report, Milestones	Codoban, Toferer
7	16.3.2026	Updating Milestone Approval, Adding Progress Report	Toferer

Distribution list

Name	Organisational Unit	Role in project	Telephone	E-mail
Susane Loidl	PAG	customer		Susane.loidl@bildung.gv.at
Thomas Toferer	PM	PM and Hardware Design		Thomas.toferer@htl-wels.at
Laurenz Durkowitch	TM	Comunictaion between Devices		Laurenz.durkowitch@htl-wels.at
Codoban Simon	TM	Setting up Hardware, Documentation		Simon.Codoban@htl-wels.at
Andreas Wögerbauer	External advisor	Advising in Hardware selection		Andreas.woegerbauer@bildung.gv.at

1 Project plans

1.1 Project Assignment

Smart-Wand 1	PROJECT-ASSIGNMENT
Project start event: • **12.1.2026**	**Project start date:** • **12.1.2026**
project close down event in terms of content: • **End of Mai** **Formal project close down event:** • **End of Mai**	**Project close down dates:** • **10.6.206**
Project objectives: • **Creating useable Wand for Smart Devices** • send commands to Home Assistant • Distinguish different Spells by Movement	**Non-objectives:** • Expanding support for other Smart-Home management Systems (Google Home, Loxone etc) • Lifetime Support • Further Development
Main tasks (Project phases): • **Detecting Spells (Gyro, Acceleration)** • Send commands to Home-Assist • Refine spell detection	**Project resources and costs*:** (see table below)
Project owner: • **HTL-Wels**	**Project manager:** • **Toferer Thomas**
Project team members: • Codoban Simon • Durkowitsch Laurenz	
HTL Wels.. (Project owner)	Toferer Thomas.. (Project manager)

resource/type of cost	unit	Costs (€)
ESP 32	1	7€
RasperryPi 5	1	75€
MPU6050	1	7,55€
Shelly Plug	1	23,8€

* Possible categories of total Project budget:
Category A: cover of the shelly module
Category B: full cover of equipment

1.2 Project Objectives (objectives, non-objectives)

Smart Wand 1	PROJECT OBJECTIVES	
Type of objective	**Project objectives**	**Adjusted project objectives as of...**
objectives: • **Main objectives** • **Additional objectives**	• Detect multiple Spells • Communicate with Home-Assist • **Create a reliable and nice experience using the device** • **Add Support for another microcontroller** • **Web interface to manage Spells**	
• **Non-objectives**	• **Add Support for other IOT Management Solutions** • **Offer Lifetime Support** • **Further Developement**	• If there is enough time left before end of mai this objective could be transformed into an additional object.

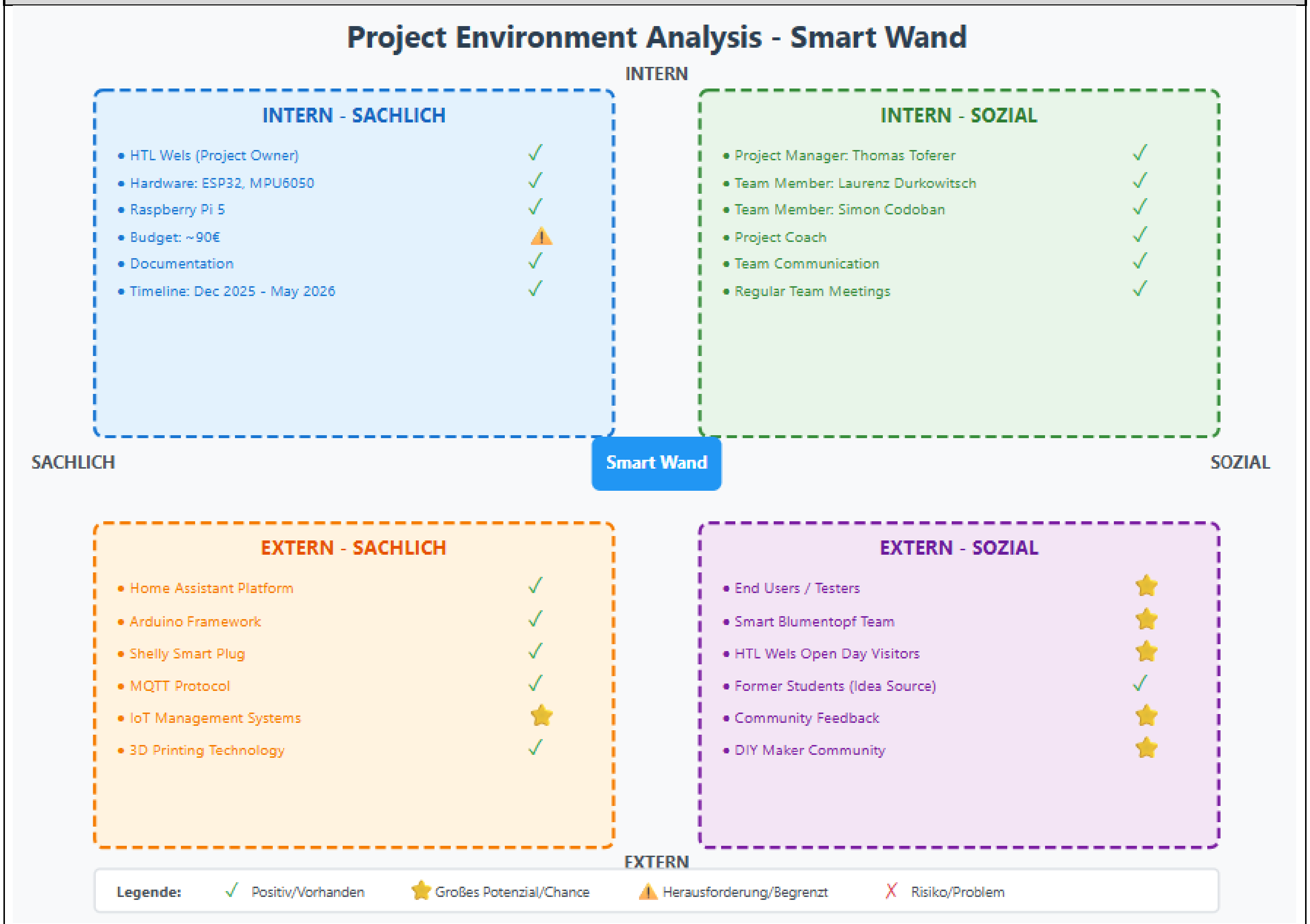
1.3 Description of Pre- and Post Project Phase

Smart Wand 1	DESCRIPTION OF PRE- AND POST-PROJECT PHASE
1) Pre-project phase	
	What triggered the project? • *The project was proposed by a school assignment in ITP.* • *The idea of the project was given to us by a former student of HTL Wels.* • *The project idea was exactly what we wanted for an educational project. To develop this accessory, we need to build skills in various fields of computer science, offering a good overview of choosing the right career path in the future.*
	Relevant documents for the project („Minutes", … ONLY documents and no content necessary) • *SmartWand_PSP* • *SmartWand_Goals* • *Requirement Spezification(Pflichtenheft)*
	Experience from similar projects • *Xdj100s build (Toferer)* • *FPV Drones (Durkowitsch)*
2) Post-project phase	
	What will happen after the project has ended? • *Demonstration of the project on "open day" of HTL Wels 2026*

1.4 Project Environment Analysis

Smart Wand 1

PROJECT ENVIRONMENT GRAPHIC



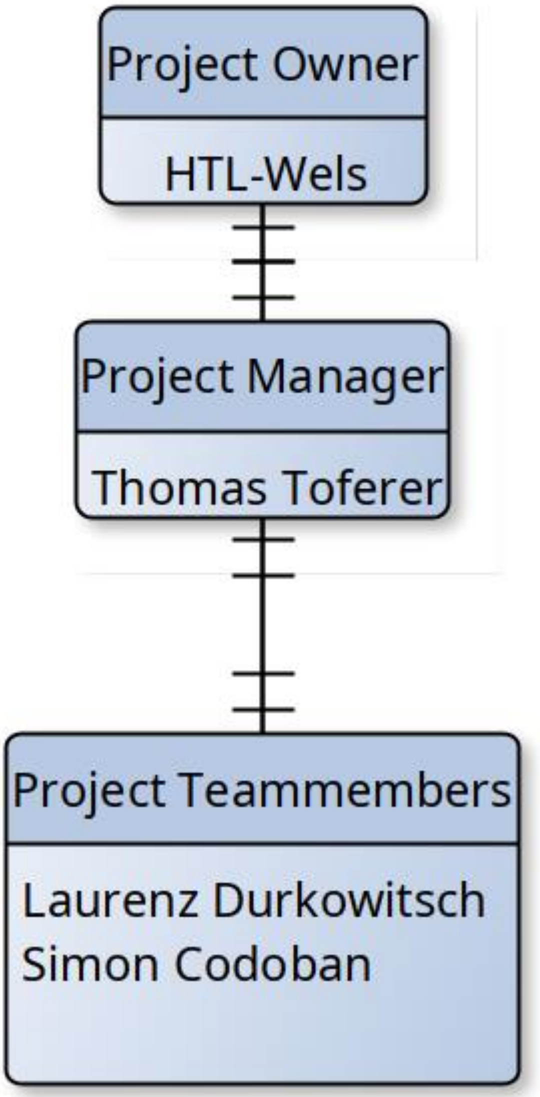
Smart Wand 1	PROJECT ENVIRONMENT TABLE		
Environment	**Relationship** (potential/conflict)	**Measures**	**Who / when** **PSP Code**
Internal	Investor	**Regular updates on project progress**	**HTL Wels alltime**
Tester/User	Feedback, Finding Bugs	Requirements gathering, early feedback	Everybody, Demo 1.1.1.4

1.5 Relationship to Other Projects and the Organisations´s Strategy

Smart Wand 1	RELATIONSHIP TO OTHER PROJECTS		
Programs/ Projects/	**Relationship** (potential/conflict)	**Sanctions**	**Who / when** **WBS Code**
Arduino Framework	We need those Open-Source Libraries to get sensor data. (get prohibited)	**Development will be more difficult**	1.1
Home Assistant	Managing IOT Devices and REST API calls. (get prohibited)	The project will have to use another IOT Management-Software	
Smart Blumentopf	Management Cooperation. Argument between PMs	Similar problems both projects could have, can't be solved with their knowledge.	

Smart Wand 1	CONNECTION TO THE ORGANISATION´S STRATEGY
Strategy	**Description of connection/relationship**
-	-

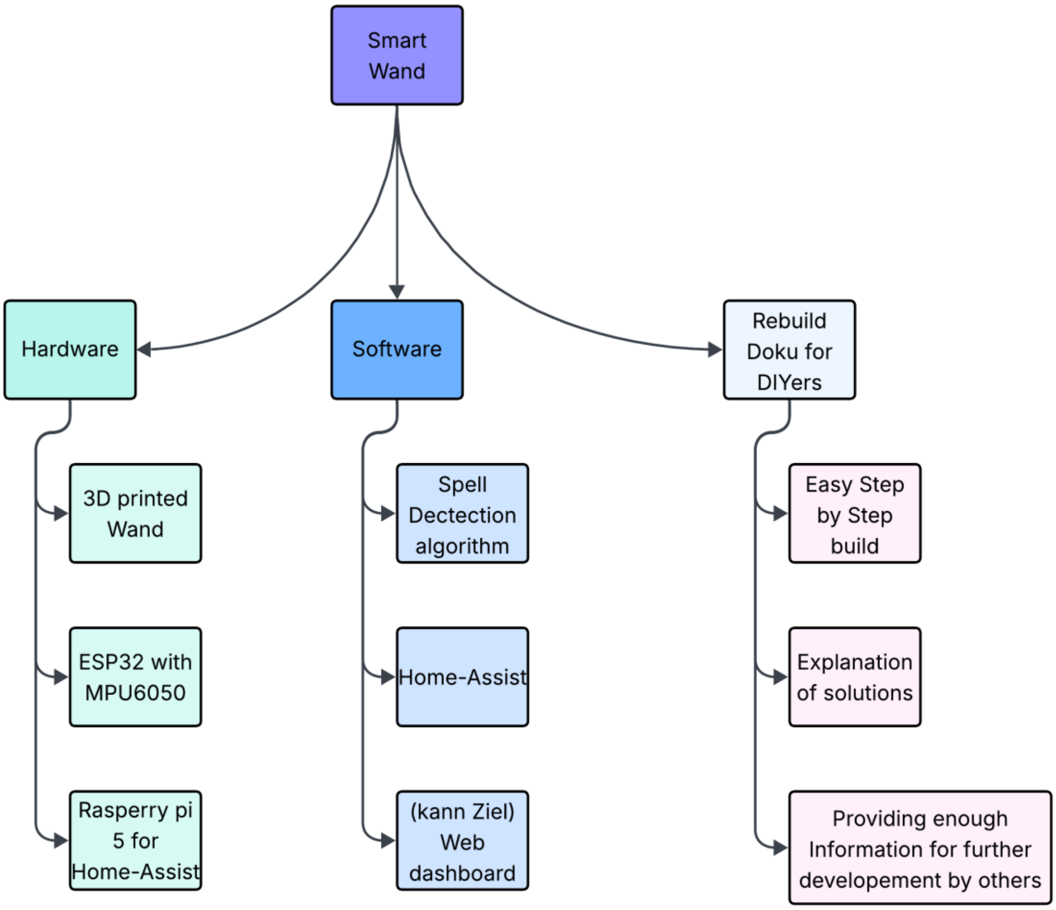
1.6 Project Organisation Chart



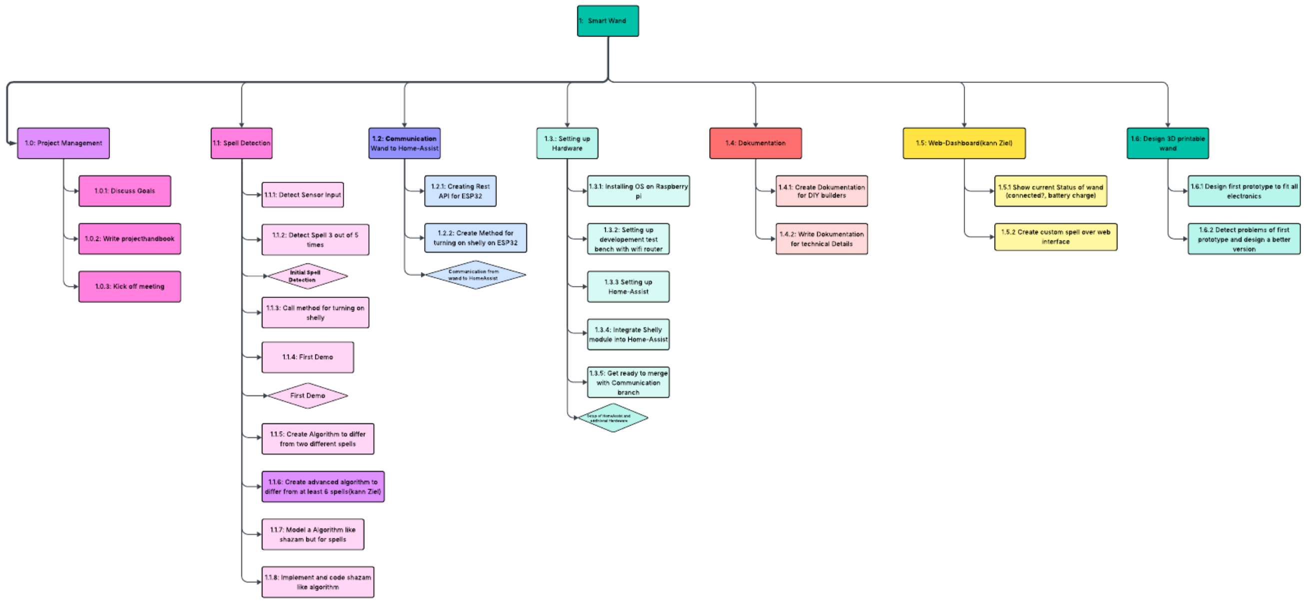
Smart Wand 1	PROJECT- ORGANISATION	
Role in Project	**Field of duties/Skills**	**Name**
Project owner	Advisor for technical problems. Investor	HTL Wels
Project manager	Coordination, Development of Spell Detection	Toferer Thomas
Project team members	1. Development of Rest API 2. Helping with smaller tasks	1. Laurenz Durkowitsch 2. Simon Codoban
Project members		

1.7 Plan of Objects of Consideration of the Project

Smart Wand 1	PLAN OF OBJECTS OF CONSIDERATION



1.8 Work Breakdown Structure (WBS)



Simple org chart.pdf

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1.9 Project Work-Package Specification

Smart Wand 1	PROJECT WORK-PACKAGE SPECIFICATION

1.1 Spell Detection

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.1.1 Detect Sensor Input	Implement sensor data collection from wand components Establish baseline for movement detection	Process sensor data for spell recognition Implement data filtering algorithms	Reliable sensor input detection system	Sensor data visible in development environment Consistent data stream verified for 5 minutes
1.1.2 Detect Single Spell 4 out of 5 times	Develop algorithm to recognize one specific spell movement Test and refine recognition accuracy	Implement multiple spell detection Add voice recognition capabilities	Reliable single spell detection system	Detection of spell 4 out of 5 times performing the movement
1.1.3 Call method for turning on Shelly	Implement API call to trigger Shelly device Integrate spell detection with device control	Create new Shelly devices Modify Shelly firmware	Successful triggering of Shelly device via spell	Shelly device responds to spell detection 4 out of 5 times
1.1.4 First Demo	Prepare demonstration of basic spell detection Showcase wand movement to device control functionality	Include multiple spells in the demo Demonstrate advanced features not yet implemented	Working demo showing basic functionality	Successful demo presented to project stakeholders Basic spell detection and device control demonstrated
1.1.5 Create Algorithm to differ from two different spells	Develop algorithm to distinguish between two different spell	Implement more than two spell recognition	System that can reliably differentiate	Correct identification of both spells 4 out of 5 times

	movements Implement classification logic for multiple gestures	Add machine learning components	between two spells	
1.1.6 Create code to collect training data	Collect training data from ESP by recording different movements	The data shall not be collected	Data can be collected efficiently	Recording Data makes no problems
1.1.7 Collect training data	Cast spells to get data for AI training	AI training and data cleaning	There are enough samples recorded to start AI training	Model documented and validated with sample data Concept approved by technical lead
1.1.8 Clean Data for AI training	Data collected needs to be cleaned of noise and further distractions	AI training	Data can be used for training	The data is analyzed and clean
1.1.9	AI Training	AI implementation	AI can detect spells	AI can recognize and detect spells 4 out of 6 times

1.2 Communication

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.2.1 Creating Rest API for ESP32	Develop REST API interface for ESP32 microcontroller Enable communication between wand and central system	Implement MQTT protocol in this phase Create web interface for API testing	Functional REST API on ESP32	API responds to basic HTTP requests Successful data transmission verified
1.2.2 Create Method for turning on shelly on ESP32	Implement specific API endpoint for Shelly device control	Modify Shelly device firmware Implement multiple	Reliable method to control Shelly devices via API	Shelly device responds to API commands 4 out of 5 times Command

	Create command structure for device activation	device control		structure documented

1.3 Setting up Hardware

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.3.1 Installing OS on Raspberry pi	Install operating system on Raspberry Pi Configure basic system settings	Install application-specific software Configure network services	Functional Raspberry Pi with operational OS	OS successfully boots Basic system commands work
1.3.2 Setting up development test bench with wifi router	Configure isolated network environment for testing Set up necessary development tools	Configure production network settings Install final application software	Ready-to-use development environment	Test bench operational with network connectivity All team members can access the environment
1.3.3 Setting up Home-Assist	Install and configure Home Assistant platform Set up basic system structure	Integrate with Shelly devices yet Configure advanced automation	Operational Home Assistant instance	Home Assistant accessible via web interface Basic configuration completed
1.3.4 Integrate Shelly module into Home-Assist	Add Shelly device integration to Home Assistant Configure basic device control	Create complex automations Modify Shelly device settings directly	Shelly devices controllable through Home Assistant	Shelly devices appear in Home Assistant interface Basic on/off control working
1.3.5 Get ready to merge with Communication branch	Prepare hardware configuration for integration Document	Implement communication features Modify existing	Hardware setup ready for integration with communication system	Hardware configuration documented Readiness confirmed by both

	current setup for team reference	communication code		hardware and communication teams

1.4 Documentation

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.4.1 Create Documentation for DIY builders	Document all steps required for DIY builders to replicate the project Create clear, step-by-step instructions	Include advanced troubleshooting beyond basic setup Document internal development processes	Complete DIY builder guide	Guide covers all necessary steps for project replication Guide validated by external tester
1.4.2 Write Documentation for technical Details	Document technical specifications and implementation details Create reference material for developers	Include user-facing instructions Document features not yet implemented	Comprehensive technical documentation	All technical components documented Documentation reviewed and approved by technical lead

1.5 Web-Dashboard

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.5.1 Show current Status of wand	Display wand connection status Show battery percentage information Present connection status information	Allow spell triggering via dashboard Include advanced configuration options	Dashboard showing all key wand status information	All status parameters displayed correctly Information updates in real-time
1.5.2 Create custom spell over web interface	Implement interface for creating new spell patterns	Allow real-time spell execution from dashboard	Functional interface for custom spell creation	Users can create and save at least one custom spell Spell data

	Allow users to define and save custom spells	Implement advanced spell editing features		correctly transmitted to wand system

1.6 Design of the Wand (3D Model)

Work Package	WP Content (What shall be done?)	Non-WP Content (What shall not be done?)	WP Result (What is achieved after WP was finished?)	Progress Measurement (How is progress measured?)
1.6.1 Design first prototype to fit all electronics	Create 3D model of wand with space for all components Ensure ergonomic design while accommodating electronics	Create final production-ready design Optimize for mass manufacturing	First functional prototype design	3D model completed with all components fitting Design reviewed and approved by team
1.6.2 Detect problems of first prototype and design a better version	Test first prototype and identify issues Create improved design based on testing feedback	Create multiple design variations Finalize for production without further testing	Improved wand design addressing prototype issues	Issues from first prototype documented Improved design created and validated with component fit test

1.10 Project Responsibility Matrix

SmartWand 1

PROJECT-RESPONSIBILITY-MATRIX

WBS-Code	WP-Title / Roles & Environment	External project owner	Project owner	Project manager	Project assistant	Project team member 1	Project team member 2	Project team member 3	Project team member 4	Project team member 5	Project member 1	Project member 2	Project member 3	Project coach
1.0	**Project Management**		**I**	**R**		**C**	**C**							
1.0.1	**Set goals for project**		**I**	**R**		**C**	**C**							
1.0.2	**Write Project-Handbook**		**I**	**R**		**C**	**C**							
1.0.3	**Kick-off meeting**		**I**	**R**		**C**	**C**							
1.1	**Spell Detection**		**I**	**R**		**C**	**I**							
1.1.1	First spell detection		I	R		C	I							
1.1.1.1	Detect Sensor Input		I	R		I	I							
1.1.1.2	Detect spell 4 out of 5 time		I	R		I	I							
1.1.1.3	Call method for turning on shelly		I	R		C	I							
1.1.1.4	First Demo		I	R		C	C							
1.1.1.5	Create Algorithm for 2 spells		I	R		I	I							
1.1.2	Advanced Algorithm		I	R		I	C							
1.1.2.1	Model a Algorithm like shazam but for spells		I	R		I	I							
1.1.2.2	Implement and code shazam like Algorithm		I	R		I	I							
1.2	**Communication**		**I**	**I**		**R**	**I**							
1.2.1	Creating Rest API for ESP32		I	I		R	I							
1.2.2	Create Method for turning shelly on ESP32		I	I		R	I							
1.3	**Setting up Hardware**		**I**	**I**		**R**	**C**							
1.3.1	Installing OS on Raspberry pi		I	I		R	C							
1.3.2	Setteing up developement-bench		I	I		I	R							
1.3.3	Setting up Home-Assist		I	I		R	C							
1.3.4	Integrate Shelly module into Home-Assist		I	R		C	R							
1.3.5	Get ready to accept calls from ESP		I	I		R	C							
1.4	**Dokumentation**		**I**	**C**		**I**	**R**							
1.4.1	Create Docu for rebuilders		I	I		C	R							
1.4.2	Write technical documentation		I	C		C	R							
1.5	**Webdashboard**		**I**	**I**		**R**	**C**							
1.5.1	Show current status of wand		I	I		R	C							

1.5.2	Create custom spells over webdashboard		I	C		R	C							
1.6	**Design wand**		**I**	**C**		**R**	**I**							
1.6.1	Design first prototype		I	C		R	I							
1.6.2	Redesign first design		I	C		R	I							

Functions
R …….Responsible
C …….Contribution
I ……..has to be informed

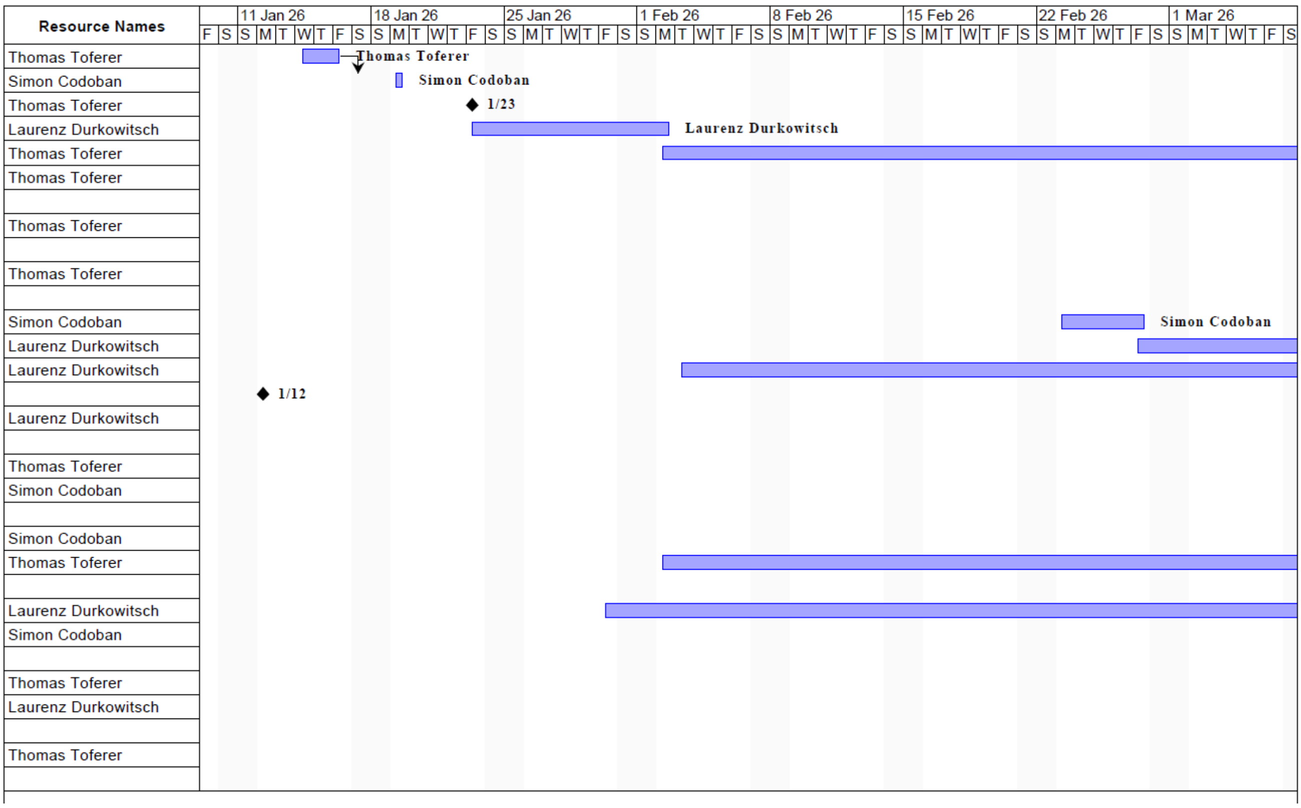
1.11 Milestoneplan

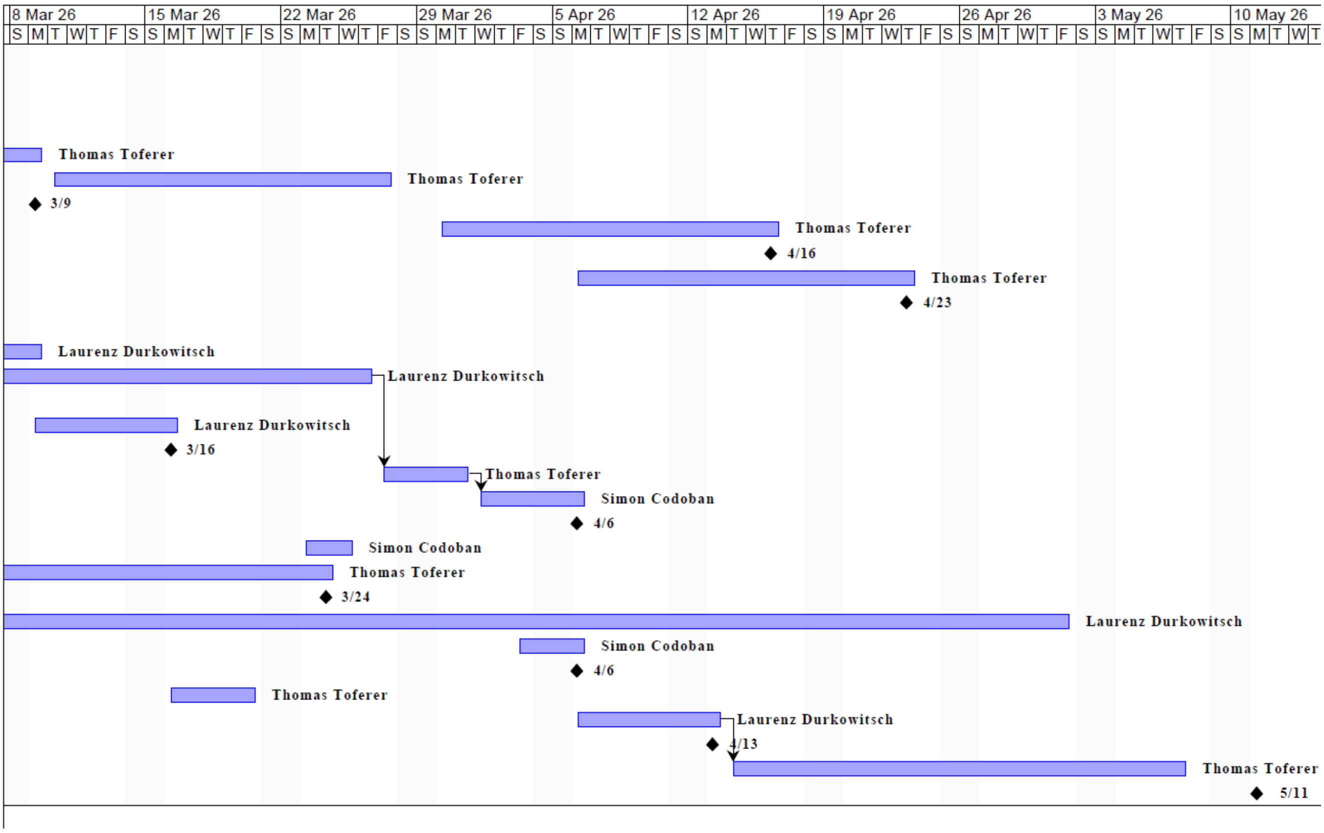
<Project name> <Project no.>	MILESTONEPLAN			
WBS-Code	**Milestone**	**Plan date**	**Revised date**	**Actual date**
1.1.2	Initial Spell Detection	7 February		2.2.2026
1.3	Setup of HomeAssist and addinional Hardeware	11 February		5.2.2026
1.2	Communication from wand to HomeAssist	18 March		
1.1.5	Training Data Collected	25 March		
1.1.4	Preperation for first Demo	8 April		
1.5	Web Dashboard first version functional	15 April		
1.1.6	AI Trained	1 May		
1.6.1	3D Model of wand finalized	10 May		
1.1.7	AI model documented	20 May		
1.4	Complete documentation ready	5 June		
	Final Demo and Project Presentation	9 June		

*In order of plan dates.

1.12 Project Bar Chart

		Name	Duration	Start	Finish	Predecessors
1		1.1.1 Detect Sensor Input	2 days	1/14/26, 9:00AM	1/16/26, 9:00AM	
2		1.1.2 Detect Single Spell 4 out of 5 times	1 day	1/17/26, 8:00AM	1/19/26, 5:00PM	1
3		1.1.4 First Demo	0 days	1/23/26, 8:00AM	1/23/26, 8:00AM	
4		1.1.5 Create Algorithm to differ from two different spells	7 days?	1/23/26, 8:00AM	2/2/26, 5:00PM	
5		1.1.6 Create Code to gather training data	26 days?	2/2/26, 8:00AM	3/9/26, 5:00PM	
6		1.1.7 Collect trainings data	14 days?	3/10/26, 8:00AM	3/27/26, 5:00PM	
7		1.1.7.7 Enough training data to continue	0 days	3/9/26, 8:00AM	3/9/26, 8:00AM	
8		1.1.8 Preparing data for training	14 days?	3/30/26, 7:00AM	4/16/26, 5:00PM	
9		1.1.8.8 Data is prepared for training	0 days	4/16/26, 7:00AM	4/16/26, 8:00AM	
10		1.1.9 Start training AI	14 days	4/6/26, 7:00AM	4/23/26, 5:00PM	
11		1.1.9.9 AI training is running	0 days	4/23/26, 7:00AM	4/23/26, 8:00AM	
12		1.2.1 Creating Rest API for ESP32	5 days?	2/23/26, 8:00AM	2/27/26, 5:00PM	
13		1.2.2 Create Method for turning on shelly on ESP32	7 days	2/27/26, 8:00AM	3/9/26, 5:00PM	
14		1.3.1 Installing OS on Raspberry pi	38 days	2/3/26, 8:00AM	3/26/26, 5:00PM	
15		1.3.1.1 The OS is running without bigger problems	0 days	1/12/26, 8:00AM	1/12/26, 8:00AM	
16		1.3.2 Setting up development test bench with wifi router	6 days?	3/9/26, 8:00AM	3/16/26, 5:00PM	
17		1.3.2.2 Working test bench, all Hardware set up	0 days	3/16/26, 8:00AM	3/16/26, 8:00AM	
18		1.3.3 Setting up Home-Assist	3 days?	3/27/26, 8:00AM	3/31/26, 5:00PM	14
19		1.3.4 Integrate Shelly module into Home-Assist	4 days?	4/1/26, 8:00AM	4/6/26, 5:00PM	18
20		1.3.4.4 Having a Working Home Assistant where Shelly can..	0 days	4/6/26, 7:00AM	4/6/26, 8:00AM	
21		1.3.5 Get ready to merge with Communication branch	3 days?	3/23/26, 8:00AM	3/25/26, 5:00PM	
22		1.4.1 Create Documentation for DIY builders	37 days	2/2/26, 8:00AM	3/24/26, 5:00PM	
23		1.4.1.1 Finished Doc, maybe feedback from the community	0 days	3/24/26, 8:00AM	3/24/26, 8:00AM	
24		1.4.2 Write Documentation for technical Details	66 days?	1/30/26, 7:00AM	5/1/26, 5:00PM	
25		1.5.1 Show current Status of wand	2 days?	4/3/26, 7:00AM	4/6/26, 5:00PM	
26		1.5.1.1 Had a good presentation of you wand	0 days	4/6/26, 7:00AM	4/6/26, 8:00AM	
27		1.5.2 Create custom spell over web interface	5 days?	3/16/26, 8:00AM	3/20/26, 5:00PM	
28		1.6.1 Design first prototype to fit all electronics	6 days?	4/6/26, 7:00AM	4/13/26, 5:00PM	
29		1.6.1.1 Received and tested the first prototype	0 days	4/13/26, 7:00AM	4/13/26, 8:00AM	
30		1.6.2 Detect problems of first prototype and design a bette..	18 days?	4/14/26, 8:00AM	5/7/26, 5:00PM	28
31		End of Project	0 days	5/9/26, 7:00AM	5/11/26, 5:00PM	





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1.13 Resource Plan

<Project name> <Project no.>			Resource Plan			
WBS-Code	**Phase/Work-package**	**Type of resource**	**Planned quantity (days)**	**Revised quantity (days)**	**Actual quantity (days)**	**Deviation (days)**
	Detecting spells	**Time, ESP32, MPU6050**	**30**			
	Developeping Rest API	Time	30			
	Configure Hardeware wth Home-Assist	Time, Rasperry pi,	15			
	Writing Dokumentation	Time	30			

1.14 Project Cost Plan

Smart Wand 1	PROJECT COST PLAN				
WBS-Code, WP-Title	Type of Cost	Planned cost	Revised cost	Actual cost	Deviation
	• Personnel	0	0	0	
	• Material	< 90€			
	• External services	0	0	0	
	• Other	0	0	0	
	Total	< 90€			
	• Personnel				
	• Material				
	• External services				
	• Other				
	Total				
	• Personnel				
	• Material				
	• External services				
	• Other				
	Total				
Project cost					

1.15 Project Communication

<Smart Wand> <1>	PROJECT- COMMUNICATION			
Title	**Objectives, Content**	**Participants**	**Schedule**	**Location**
Project owner meeting	• project status • decisions • acceptance of progress report	Project owner, Project manager		
Project controlling meeting	• project status • controlling of tasks, schedule, resources, costs • controlling of project environments • social controlling • prepare proposal for decision	Project manager, Project team, Project coach		
Subteam meeting	• Coordination of subteams • Discussion of problems	Subteam		
	•			

1.16 Project „Rules"

1.17 Project Risk Analysis

Risk Analysis completed on: January 10, 2025

PROJECT RISK ANALYSIS									
WBS-Code (Code)	WBS-description (Text)	Risk-description (Text)	Priority (Range)	Benefit /Cost (iuro)	Probability (Percentage)	Risk-value (iuro)	Delay (iWeeks)	Avoiding action (Text)	Cost for minimizing risk (Euro)
1.1	Spell Detection	Sensor calibration issues leading to inaccurate spell detection	High	€0	40%	€0	2 weeks	Regular calibration tests, backup sensors available	€50
1.2	Communi cation	MQTT connection failures between wand and Home Assistant	Medi um	€0	30%	€0	1 week	Implement connection retry logic, test network stability	€0
1.3	Hardware Setup	Raspberry Pi hardware failure or unavailability	Medi um	€40	20%	€8	1 week	Have backup Raspberry Pi ready, regular backups	€40
1.1.8, 1.1.9	AI training	AI doesn’t work as expekted	High	€0	35%	€0	3 weeks	Optimize code early, consider ESP32-S3 upgrade if needed	€15
1.4	Documen tation	Incomplete or delayed documentation affecting project handover	Low	€0	25%	€0	-	Weekly documentation reviews, assign dedicated time slots	€0
1.6	3D Design	3D printing failures or design flaws requiring redesign	Medi um	€5	30%	€6	1 week	Test print prototypes early, multiple design iterations	€20

1.18 Project Documentation

Area	Description
File	Technical_Documentation_Smart-Wand.pdf Rebuild_Guide_Smart-Wand.pdf
Access Authorisation	All Project members Main responsibility: Codoban Simon
Naming convention	Versioning rules (v1.0, v1.1, Final, etc.) ProjectName_DocumentType_Version_V1.0.pdf
Rules	Update responsibility Documentation must be versioned via Git The current version must be uploaded to GitHub After each Milestone the Documentation shall be updated

2 Project Start

2.1 Minutes– Project Start

2.1.1 Project Start Workshop

Questions to be discused:

- Who is responsible for what?
- What are the working hours? During
- When will the Team-Building-Events happen?
- Where will the Team-Building-Events happen? Kaffeeautomat?

2.1.2 Follow-up Workshop

In the follow up meeting which will happen on 1.3.2026 the current conventions will be discussed and if necessary, changes will be made.

2.1.3 Project Owner Meeting

As of now there is no Project Owner meeting planned

3 Project Co-ordination

3.1 Approval of Work-packages

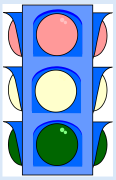
<Smart Wand> <1> APPROVAL OF WORK-PACKAGES					
WBS-Code	**Work-package**	**WP-Owner**	**Date**	**Approval by**	**Signature**
1.3.1	Installing OS on Raspberry pi	Durkowitsch	3.2.2026	Toferer	*Toferer*
1.1.1	Detect Sensor Input	Toferer	10.2.2026	Toferer	*Toferer*
1.1.2	Detect Single Spell 4 out of 5 times	Codoban		Toferer	
1.1.3	Call method for turning on Shelly	Toferer	18.3	Toferer	*Toferer*
1.1.4	First Demo	Toferer		Toferer	
1.1.5	Create Algorithm to differ from two different spells	Durkowtisch		Toferer	
1.1.6	Create code to collect training data	Codoban	9.3.2026	Toferer	*Toferer*
1.1.7	Collect Training data	Durkowitsch		Toferer	
1.1.8	Clean Training data	Toferer		Toferer	
1.1.9	Train AI	Toferer			
1.2.1	Creating Rest API for ESP32	Codoban		Toferer	
1.2.2	Create Method for turning on shelly on ESP32	Durkowitsch		Toferer	
1.3.2	Setting up development test bench with wifi router	Durkowitsch	20.2.2026	Toferer	*Toferer*
1.3.3	Setting up Home-Assist	Toferer	27.2.2026	Toferer	*Toferer*
1.3.4	Integrate Shelly module into Home-Assist	Codoban	29.2.2026	Toferer	*Toferer*

1.3.5	Get ready to merge with Communication branch	Codoban		Toferer	
1.4.1	Create Documentation for DIY builders	Toferer		Toferer	
1.4.2	Write Documentation for technical Details	Durkowitsch		Toferer	
1.5.1	Show current Status of wand	Codoban		Toferer	
1.5.2	Create custom spell over web interface	Toferer		Toferer	
1.6.1	Design first prototype to fit all electronics	Durkowitsch		Toferer	
1.6.2	Detect problems of first prototype and design a better version	Toferer		Toferer	

3.2 Minutes – Project Co-ordination

4 Project Controlling

4.1 Project Status Report

<Smart Wand> <1>	PROJECT STATUS REPORT as of 3.2.2026 by Thomas Toferer
 Project crisis Project in difficulties Project according to plan ✓	**0Overall Status:** • **Running** • **Communication to HA not yet ready**
2) Status Project objectives • **Objectives have not changed** •	**Activities:**
3) Status Project progress • **Spell detection first prototype** • **Raspberry setup almost finished**	**Activities:** • **Spell Detection** • **Setting up Home Assistant** • **Installing MQTT Broker**
4) Status Schedule • **WPs Still in time** •	**Activities:** • **Setting up Home Assistant still has some time till it must be completed**
5) Status Resources/costs • **According to plan** •	**Activities:** • •
6) Status Context • **No external dependencies currently blocking development** • **All workers are physically present**	**Activities:** • •
7) Status Organisation/culture • **All Workers are in moderate condition** •	**Activities:** • •

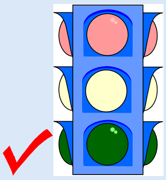
5 Project Controlling

5.1 Project Status Report

<Smart Wand>
<1>

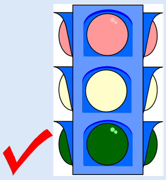
PROJECT STATUS REPORT

as of 16.2.2026 by Laurenz Durkowitsch

Project crisis Project in difficulties Project according to plan 	**1Overall Status:** • **Running** • **Request to HA via ESP32 more difficult then expected**
2) Status Project objectives • **Objectives have not changed**	**Activities:**
3) Status Project progress • **Raspberry with HomeAssistant fully setuped** • **EPS32 for request to HA in development**	**Activities:** • **MQTT installed and set up** • **HA made ready** • **Tested the project in its testnetwork** •
4) Status Schedule • **Still in time**	**Activities:** •
5) Status Resources/costs • **According to plan**	**Activities:** •
6) Status Context • **Holidays are blocking futher development until next week** • **All workers are in Holidays**	**Activities:** • •
7) Status Organisation/culture • **All Workers are still in moderate condition**	**Activities:** • •

6 Project Controlling

6.1 Project Status Report

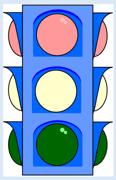
<Smart Wand> <1>	PROJECT STATUS REPORT as of 09.3.2026 by Simon Codoban
Project crisis Project in difficulties Project according to plan 	**2Overall Status:** • **Running** • **Request to HA via ESP32 more difficult then expected**
2) Status Project objectives • **Spell will now be detected using an AI**	**Activities:**
3) Status Project progress • **Project Handbook updated based on teachers feedback** • **Risk Analysis table completed (6 risks identified)** • **Milestone Plan expanded (7 new milestones added)** • **Resources names added to Gantt** • **Code to collect traning data for AI was developed** • **Project Bar Chart added**	**Activities:** • **Better design refinement** • **Training data can now be recorded**
4) Status Schedule • **Still in time**	**Activities:** •
5) Status Resources/costs • **According to plan**	**Activities:** •
6) Status Context	**Activities:** • •
7) Status Organisation/culture • **Handbook documentation improved and updated** • **All changes highlighted in yellow as required**	**Activities:** • •

6.2 Project Status Report

<Smart Wand>
<1>

PROJECT STATUS REPORT

as of 09.3.2026 by Simon Codoban

Project crisis Project in difficulties Project according to plan  ✓	**3Overall Status:** • **Running** • **Overall progress is delayed by Wand Design**

2) Status Project objectives • **Objectives have not changed**	**Activities:**
3) Status Project progress • **Wand is being designed and printed** • **Collection training data**	**Activities:** • **Better design refinement** • **Data is being recorded for the different spells** • **Communication_between HomeAssist and ESP**
4) Status Schedule • **Training data Collection is taking longer than expected but we are still in time** **"WOverall" Still in time** • **Communication between ESP and HA will only be partly finished by 18.3.**	**Activities:** • **Collection Training Data** • **Wand Design has a higher Priority than the ESP-HA Communication. Without the wand design training data can't be recorded reliable.**
5) Status Resources/costs • **According to plan**	**Activities:** •
6) Status Context	**Activities:** • •
7) Status Organisation/culture • **Handbook documentation improved and updated** • **All changes highlighted in yellow as required**	**Activities:** • •

6.3 Additional Project Status Reports

6.4 Minutes – Project controlling

6.4.1 Project Controlling Meetings

Not happening at

6.4.2 Project Owner Meetings

As of now there hasn't been any agreed dates for a Project Owner Meetings.

7 Project Close Down

7.1 Project Close Down Report

<table>
<tr><td colspan="2"><Project name>
<Project no.>

PROJECT CLOSE DOWN REPORT</td></tr>
<tr><td>1) Overall impression</td><td>2) Reflection: Fulfilment of objectives</td></tr>
<tr><td colspan="2">3) Reflection: Deliverables / Schedule</td></tr>
<tr><td colspan="2">4) Reflection: Resources / Costs</td></tr>
<tr><td colspan="2">5) Reflection: Internal Organisation / Environmental Relationships</td></tr>
<tr><td>6) Performance appraisal (Project owner, Project manager, Project member)</td><td>7) Lessons learned (Summary of Experiences and suggestions for improvement)</td></tr>
<tr><td colspan="2">8) Post-Project Phase Planning, Additional Tasks</td></tr>
</table>

To-Do	Owner	Schedule

9) Project Close Down

-- *<Name>* (Project owner)

-- *<Name>* (Project manager)

7.2 Minutes – Project Close Down

7.2.1 Project Close Down Workshop